

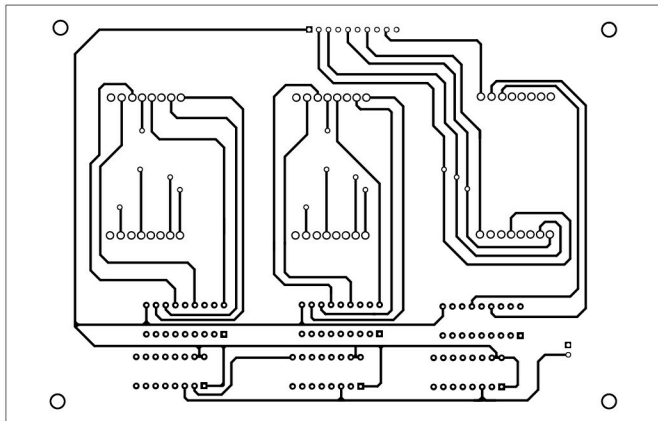


Fig. 8: Track layout of the top layer of display unit

for high-speed operation in cascaded system. The same serial information, available at the Q terminals on the next negative clock edge, provides a means for cascading CD4094 devices when the clock rise time is slow.

ULN2803 (IC6-IC8) is an octal high-voltage, high-current Darlington transistor array. The eight npn Darlington connected transistors (T1-T8) are ideally suited for interfacing between low-logic-level digital circuitry and the higher current/voltage requirements of lamps and relays. The device features open-collector outputs and freewheeling clamp diodes for transient suppression. It is designed to be compatible with standard TTL families of ICs.

Circuit and working

The circuit diagram is divided here into two parts. The first part has a power supply and controller circuit as shown in Fig. 3. The display unit is in the second part as shown in Fig. 4.

The power supply circuit is built

around a step-down transformer, bridge rectifier and 5V regulator. The configuration is conventional. The circuit provides regulated 5V for operation of the circuit.

Microcontroller AT89C52 provides outputs to control the logic levels for the eight rows R0-R7 through port A (P0.0-P0.7). RNW1 is a network resistor, which acts as a pull-up resistor for port 0. An array of eight pnp BC558 transistors (T1-T8) working as current drivers takes care of the current required for LEDs of the dot-matrix; the microcontroller has low sourcing capabilities. The reset arrangement is made at pin 9 of the microcontroller with the help of capacitor C3, resistor R2 and switch S1.

The circuit for the display unit is shown in Fig. 4. It is built around three CD4094 cascaded 8-bit shift registers (IC3-IC5), three ULN2803 high-current Darlington transistor arrays (IC6-IC8) and three 8x8 dot-matrix displays (DIS1-DIS3). Clock and data pulses are generated by the microcontroller at

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2	- AT89C52 microcontroller
IC3-IC5	- CD4094 shift register
IC6-IC8	- ULN2803 Darlington transistors array
T1-T8	- BC558 pnp transistor
LED1	- 5mm LED
BR1	- Bridge rectifier, 1A
DIS1-DIS3	- 8x8 dot-matrix display

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 680-ohm
R2-R10	- 10-kilo-ohm
RNW1	- 10-kilo-ohm network resistor

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 1 μ F, 16V electrolytic
C3	- 10 μ F, 16V electrolytic
C4-C5	- 33pF ceramic disk

Miscellaneous:

CON1, CON3	- 10-pin connector
CON2, CON4	- 2-pin connector
X ₁ , X ₁ 1	- 20MHz crystal
X1	- 230V AC primary to 0-12V, 1A secondary transformer
S1	- Tactile switch

ports P2.1 and P2.0, respectively. These are transferred to the display unit through connectors CON2 and CON4.

Clock pulses are fed to pin 3 of all